

A Symplectic N-Body Engine: Direct and Barnes–Hut Solvers Validated Against JPL Horizons and MW–M31 Dynamics

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Abstract

We present a high-performance N-body gravitational simulator targeting two distinct dynamical regimes: precision solar system integration and galactic-scale dynamics. A 4th-order Yoshida symplectic integrator is paired with two interchangeable force kernels: a SIMD- and OpenMP-vectorized direct $\mathcal{O}(N^2)$ summation, and a Barnes–Hut octree that opens cells by geometric center and accumulates forces about the center of mass, yielding leading-order quadrupole error. The solar system configuration integrates 35 bodies (the Sun, 8 planets, Pluto, and 25 natural satellites) over timescales of up to 249 years and achieves energy conservation $\Delta E/E \leq 1.5 \times 10^{-12}$. Positional accuracy against the NASA JPL Horizons DE441 ephemeris averages 0.079% (per-body maximum relative error, mean across 34 non-Sun bodies). The Barnes–Hut kernel is exercised on a Milky Way–Andromeda demonstration: two Plummer spheres initialized per van der Marel et al.

(2012), integrated at $N = 50,000$ over 6 Gyr, reproduce the predicted first close passage at 3.72 Gyr against the literature value of 3.9 Gyr (4.6% deviation, within published uncertainty on the M31 tangential velocity). A scaling benchmark on a 12-thread Zen 4 host shows a $\sim 52.6\times$ Barnes–Hut speedup over the parallel direct kernel at $N = 131,072$, with the tree winning at all measured $N \geq 512$.

Keywords: N-body simulation, symplectic integration, Yoshida integrator, Barnes–Hut, solar system dynamics, galactic dynamics, JPL Horizons

1. Introduction

The gravitational N-body problem, that is, computing the trajectories of N mutually interacting masses, is one of the oldest and most fundamental problems in computational physics. Two distinct numerical challenges arise depending on the dynamical regime. Long-duration integration of planetary orbits requires geometric integrators that preserve the symplectic structure of Hamilton’s equations, since non-symplectic schemes exhibit secular energy drift that corrupts orbital elements over extended timescales [2]. Large- N collisionless systems, by con-

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trast, exceed the budget of direct pairwise summation and require hierarchical force solvers such as the Barnes–Hut octree [5], which reduces per-step cost from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$ at the price of a controlled approximation error.

This work presents a single integration framework that addresses both regimes. The simulator supports a 4th-order Yoshida symplectic integrator [1] with two interchangeable force kernels: a vectorized direct summation for high-precision low- N work, and a Barnes–Hut octree with quadrupole-leading multipole acceptance criterion for large- N collisionless systems. We validate the framework in two regimes against independent reference data. For the solar system, the Newtonian 35-body configuration is compared against the NASA JPL Horizons DE441 ephemeris, which incorporates post-Newtonian relativity, solar oblateness, asteroid perturbations from over 300 bodies, and tidal effects [3]. For galactic dynamics, a Milky Way–Andromeda Plummer collision is compared against the first-passage prediction of van der Marel et al. [11].

The principal implementation and validation results of this work are:

1. A 4th-order Yoshida symplectic integrator achieving $\Delta E/E \leq 10^{-12}$ over multi-century solar system timescales.
2. A Barnes–Hut octree force kernel with an explicit leaf discriminant and a multipole acceptance criterion that uses geometric box center for separation and center of mass for force accumulation, yielding $\sim 53\times$ speedup over the parallel direct kernel at $N = 131,072$.
3. Positional validation against JPL DE441 for 35 solar system bodies, with quantitative attribution of residual errors to specific missing physics.
4. A Milky Way–Andromeda demonstration with van der Marel et al. (2012) initial conditions, reproducing first-passage timing to 4.6% accuracy at $N = 50,000$ over 6 Gyr.
5. A unified binary initial-conditions format with runtime-configurable scenarios, removing compile-

time branching between the solar and galactic configurations.

2. Theoretical Background

2.1. Equations of Motion

The gravitational acceleration of body i due to all other $N - 1$ bodies is given by Newton’s law of universal gravitation:

$$\mathbf{a}_i = \sum_{\substack{j=1 \\ j \neq i}}^N \frac{G m_j (\mathbf{r}_j - \mathbf{r}_i)}{(|\mathbf{r}_j - \mathbf{r}_i|^2 + \varepsilon^2)^{3/2}} \quad (1)$$

where $G = 6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ is the gravitational constant, m_j is the mass of body j , and $\mathbf{r}_i$ and $\mathbf{r}_j$ are position vectors. The softening parameter ε is configured per scenario at runtime through the initial-conditions header. For solar system integration $\varepsilon = 10^{-9} \text{ m}$, which is negligible relative to inter-planetary distances and serves only as a numerical safeguard against zero separation. For the galactic configuration $\varepsilon = 100 \text{ pc}$, which is comparable to the inter-particle spacing of the macro-particle sampling and suppresses spurious two-body relaxation; see Section 3.5.2.

2.2. Symplectic Integration

Hamiltonian systems possess a symplectic structure: the phase-space flow preserves the canonical 2-form $\sum_i dp_i \wedge dq_i$. Standard Runge–Kutta methods violate this structure, causing secular energy drift proportional to integration time. Symplectic integrators, by construction, approximately preserve a nearby shadow Hamiltonian $\tilde{H} = H + \mathcal{O}(\Delta t^p)$, bounding energy error over exponentially long timescales in $1/\Delta t$ [2]. The energy error remains oscillatory and does not exhibit secular growth.

2.2.1. Velocity Verlet 2nd-Order Integrator

The Velocity Verlet algorithm [4] is a 2nd-order symplectic method widely used in molecular dynamics:

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \Delta t \mathbf{v}(t) + \frac{1}{2} \Delta t^2 \mathbf{a}(t) \quad (2)$$

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \frac{1}{2} \Delta t [\mathbf{a}(t) + \mathbf{a}(t + \Delta t)] \quad (3)$$

This requires storing the previous acceleration and one force evaluation per timestep.

2.2.2. Yoshida 4th-Order Integrator

The Yoshida method [1] constructs a 4th-order symplectic scheme by composing three leapfrog sub-steps with carefully chosen coefficients. Define:

$$w_1 = \frac{1}{2 - 2^{1/3}}, \quad w_0 = -\frac{2^{1/3}}{2 - 2^{1/3}} \quad (4)$$

The position drift and velocity kick coefficients are:

$$c_1 = c_4 = \frac{w_1}{2}, \quad d_1 = d_3 = w_1 \quad (5)$$

$$c_2 = c_3 = \frac{w_0 + w_1}{2}, \quad d_2 = w_0 \quad (6)$$

Each timestep proceeds as four position drifts interleaved with three force evaluations (“kicks”):

$$\mathbf{r} \xrightarrow{c_1} \mathbf{v} \xrightarrow{d_1} \mathbf{r} \xrightarrow{c_2} \mathbf{v} \xrightarrow{d_2} \mathbf{r} \xrightarrow{c_3} \mathbf{v} \xrightarrow{d_3} \mathbf{r} \xrightarrow{c_4} \quad (7)$$

Here $\mathbf{r} \xrightarrow{c_k}$ denotes $\mathbf{r} \leftarrow \mathbf{r} + c_k \Delta t \mathbf{v}$ and $\mathbf{v} \xrightarrow{d_k}$ denotes $\mathbf{v} \leftarrow \mathbf{v} + d_k \Delta t \mathbf{a}(\mathbf{r})$.

The scheme achieves $\mathcal{O}(\Delta t^4)$ local truncation error with only 3 force evaluations per step, compared to 4 for classical RK4. The negative coefficient $w_0 < 0$ (equivalently $d_2 < 0$) introduces a backward sub-step that cancels lower-order error terms. The symplectic structure guarantees that energy errors remain bounded and oscillatory, never exhibiting secular growth.

2.3. Barnes–Hut Tree Method

The symplectic integrators of the preceding sections are agnostic to how the force is evaluated; they only require a routine that returns $\mathbf{a}(\mathbf{r})$ given the instantane-

ous configuration. For the low- N solar system this routine is a direct $\mathcal{O}(N^2)$ summation. For collisionless systems with $N \gtrsim 10^3$, however, direct pairwise summation becomes prohibitive. The Barnes–Hut algorithm [5] reduces the per-step cost to $\mathcal{O}(N \log N)$ by recursively grouping distant particles into multipole approximations.

2.3.1. Octree construction

The simulation volume is bounded by a cubic root cell sized to enclose all particles. Each cell that contains more than one particle is subdivided into 8 octants, recursively, until each leaf contains at most a single particle or a maximum depth is reached. Each internal node stores the total mass M , the center of mass $\mathbf{R}_{\text{COM}}$, the geometric box center $\mathbf{c}_{\text{box}}$, and the box edge length s . Leaf nodes additionally record an offset and count into a contiguous particle-index array.

In our implementation each node carries an explicit boolean leaf discriminant in addition to its child pointers. This eliminates an ambiguity in which a sentinel value such as “first child index = -1 ” is overloaded to indicate both leaves and internal nodes whose first octant happens to be empty. Under non-uniform distributions, such as those of galactic discs or two-body close encounters, the overloaded sentinel admits silent subtree skips or out-of-bounds reads into the leaf index array. The explicit boolean removes the ambiguity unconditionally.

2.3.2. Multipole acceptance criterion

For each particle, the tree is traversed top-down. At each node, the multipole acceptance criterion (MAC) decides whether the node may be approximated as a single point mass or whether traversal must descend into its children. We use the standard opening criterion

$$\frac{s}{d} < \theta \quad (8)$$

where s is the node’s box edge length, $d = |\mathbf{r}_i - \mathbf{c}_{\text{box}}|$ is the distance from the target particle to the geometric box center, and θ is the opening angle parameter.

If the criterion is satisfied, the node’s gravitational contribution is computed from a single point mass located at $\mathbf{R}_{\text{COM}}$.

Two conventions on the choice of reference point for d appear in the literature: distance to the geometric box center, or distance to the center of mass. We adopt the box center. Under uneven mass distributions the COM can lie arbitrarily close to the cell boundary, so a COM-based criterion permits the opening test to accept cells in which the target particle is in fact closer to some of the contained mass than the criterion implies, degrading the multipole approximation. The geometric center is fixed by the tree structure and bounds the worst-case separation uniformly. Force accumulation, however, is performed from the COM, since the dipole moment vanishes there and the leading-order error is quadrupole, $\mathcal{O}((s/d)^2)$ per cell.

2.3.3. Aggregate error scaling

The per-cell relative force error from a single multipole acceptance scales as $\mathcal{O}((s/d)^2) = \mathcal{O}(\theta^2)$ because the dipole correction vanishes when forces are taken about the COM. The aggregate mean relative force error across particles, however, can scale more steeply than θ^2 due to cancellations between cells with symmetric mass distributions. Hernquist [6] reports empirical aggregate scaling in the range θ^3 to θ^4 depending on the distribution and the θ range. Our test suite asserts only monotonic improvement with decreasing θ , which is the truthful weaker claim; the precise exponent is distribution-dependent.

3. Implementation

3.1. Software Architecture

The simulator is implemented in C++23 with a modular object-oriented architecture. An abstract `Integrator` base class defines the integration interface, with `Yoshida` and `VelocityVerlet` as concrete implementations. An abstract `Force` base class permits multiple force models; the current implementation provides `GravityDirect` for pairwise Newtonian

summation and `GravityBarnesHut` for octree force evaluation. The `Simulation` class orchestrates the time-stepping loop, conservation diagnostics, and binary output.

Scenario configuration has been unified into a single binary initial-conditions format. The file consists of a 40-byte header (`uint64_t N`, then five `doubles`: Δt , total time, output interval, softening ε , all in SI units), followed by N records of 32-byte null-padded ASCII names and 7-double particle states ($m, x, y, z, v_x, v_y, v_z$). Both the solar system and galactic configurations are produced and consumed through this format. The compile-time scenario branching of earlier versions has been eliminated; `main.cpp` reads the run parameters from the header at startup.

3.2. Data Layout: Structure of Arrays

Particle data is stored in a Structure-of-Arrays (SoA) layout rather than the conventional Array-of-Structures (AoS). All 13 sub-arrays (3 positions, 3 velocities, 3 accelerations, 3 old accelerations, and mass) occupy a single contiguous allocation. Each sub-array is explicitly aligned to the target SIMD width (32 bytes for AVX2, 64 bytes for AVX-512) and padded to a SIMD-element boundary so that every array start is naturally aligned for vectorized loads and stores:

```
pos_x[0..stride-1], pos_y[0..stride-1], ...
```

where the stride is N rounded up to the nearest multiple of the SIMD register width in elements (e.g., 4 doubles for AVX2). This layout maximizes cache-line utilization during the inner force loop and guarantees aligned SIMD access without runtime checks.

Raw array pointers are qualified with compiler-specific `__restrict__` annotations, informing the compiler that no aliasing occurs between position, velocity, and acceleration arrays. This enables aggressive SIMD auto-vectorization of the inner loops.

3.3. Direct Force Kernel

The direct gravitational force calculation uses $\mathcal{O}(N^2)$ pairwise summation. Self-interaction ($i = j$) is eliminated with a branchless floating-point mask:

$$\text{mask}_j = \begin{cases} 0.0 & \text{if } i = j \\ 1.0 & \text{otherwise} \end{cases} \quad (9)$$

The force contribution is multiplied by this mask, replacing a conditional branch that would prevent SIMD vectorization of the inner loop. The inner loop is annotated with `#pragma omp simd` with reductions on the acceleration accumulators to enable vectorized execution. For $N \geq 350$, the outer loop is parallelized with OpenMP via `#pragma omp parallel for`.

Newton’s third law symmetry ($\mathbf{F}_{ij} = -\mathbf{F}_{ji}$) is not exploited; each body’s acceleration is computed independently by summing over all N others. This doubles the floating-point work relative to a symmetric $j > i$ loop but is a deliberate optimization tradeoff: the regular access pattern enables SIMD vectorization of the full inner loop, and the independent per-body accumulation eliminates write conflicts under thread parallelism without requiring atomic operations.

3.4. Barnes–Hut Force Kernel

The Barnes–Hut implementation follows the structure of Section 2.3. Tree construction proceeds in two phases. The first pass computes the bounding box of all N particles and allocates a flat array of nodes sized to a conservative upper bound. The second pass inserts particles into the tree recursively, subdividing cells as required up to a maximum depth (`MAX_DEPTH = 32` in the current build). Center-of-mass and total-mass aggregates are accumulated bottom-up after insertion.

Force evaluation for each target particle uses an explicit stack-based traversal of size $8 \times (\text{MAX_DEPTH} + 2)$ to bound stack usage tightly with the tree depth. Cells passing the multipole acceptance criterion of

Eq. (8) contribute a single softened-Plummer interaction from their center of mass; cells failing the criterion have their children pushed onto the traversal stack. Leaf cells contribute direct pairwise interactions over their constituent particles, which are stored contiguously in the leaf index array to maintain locality.

The traversal is reentrant: the per-call working set (the node array, the leaf index array, and the traversal stack) is held in a local `TreeCache` object passed by reference through the call chain, rather than as mutable instance state. This permits the kernel to be invoked concurrently across the Yoshida sub-steps if desired and avoids cross-call aliasing of scratch buffers.

3.5. Initial Conditions

3.5.1. Solar system from JPL Horizons

State vectors for all 35 solar system bodies are sourced from the NASA JPL Horizons system at epoch J1950.0 (January 1, 1950, 00:00 UTC). The reference frame is solar system barycentric (SSB) with the ecliptic reference plane (ICRF). Positions are given in kilometers and velocities in km s^{-1} , converted to SI units at initialization. The body catalog comprises the Sun, 8 planets (Mercury through Neptune), Pluto, and 25 natural satellites: the Moon, Phobos, Deimos, the four Galilean moons, Amalthea, 9 Saturnian moons (Mimas through Phoebe), 5 Uranian moons (Ariel through Miranda), Triton, Nereid, and Charon.

The simulation operates in the SSB frame throughout. Because all initial state vectors are sourced from JPL Horizons in this frame, the total momentum of the 35-body subset is not identically zero (the missing solar system bodies carry the complementary momentum). No barycentric correction is applied.

An automated Python pipeline (`tools/solar/fetch.py`) queries the Horizons API, retrieves state vectors at the configured output cadence, and produces the unified binary initial-conditions file for both simulation and

validation.

3.5.2. Galactic collision: Milky Way and M31

The Barnes–Hut kernel is exercised on a Milky Way–M31 (Andromeda) close-encounter scenario. The collision is initialized with the parameters of van der Marel et al. [11]: each galaxy is represented by an isotropic Plummer sphere [8] of total mass $M = 1.5 \times 10^{12} M_\odot$ and scale radius $a = 30$ kpc. The two galaxies are placed at an initial barycentric separation of 770 kpc, with a relative velocity decomposed into a radial component of 109 km s^{-1} (closing) and a tangential component of 17 km s^{-1} . Convention: the first $N/2$ particles in the IC file are the Milky Way, and the last $N/2$ are M31.

The Plummer density and potential are

$$\rho(r) = \frac{3M}{4\pi a^3} \left(1 + \frac{r^2}{a^2}\right)^{-5/2} \quad (10)$$

$$\Phi(r) = -\frac{GM}{\sqrt{r^2 + a^2}}. \quad (11)$$

Particle positions are sampled by inverting the enclosed-mass profile in radius and assigning uniformly distributed angles on the sphere. Velocities are sampled self-consistently from the Plummer distribution function [9]. In terms of the dimensionless variable $q = v/v_{\text{esc}}(r)$, where $v_{\text{esc}}^2 = -2\Phi(r)$, the local speed distribution at radius r is

$$f(q) dq \propto q^2 (1 - q^2)^{7/2} dq, \quad 0 \leq q \leq 1. \quad (12)$$

This is sampled by rejection on uniform $q \in [0, 1]$ against the bounding constant $\max_q q^2(1 - q^2)^{7/2}$, with empirical acceptance fraction near 10%. Velocity directions are sampled isotropically. The resulting per-particle phase-space distribution is in equilibrium with the static Plummer potential; no virialization pre-pass is required.

After sampling, the total center-of-mass position and velocity of the combined two-galaxy system is subtracted, so that the binary system remains centered in the simulation volume and the initial total momentum is zero by construction.

For $N = 50,000$ macro-particles, the per-particle mass is $6 \times 10^7 M_\odot$. The softening parameter is set to $\varepsilon = 100$ pc, comparable to the mean local inter-particle spacing in the Plummer core and consistent with standard practice for suppressing two-body relaxation in collisionless N-body simulations of this resolution [10].

3.6. Build System

The project uses CMake (minimum version 3.25) for cross-platform builds. Compiler optimizations (`-O3 -march=native`) and OpenMP linkage are configured automatically. The build targets GCC, Clang, and MSVC. The repository layout separates pure C++ sources under `src/`, validation and visualization scripts under `tools/solar/` and `tools/galaxy/`, and unit and benchmark tests under `tests/`.

4. Validation Methodology

Validation is performed by direct comparison of simulated trajectories against reference data, supplemented by conservation diagnostics. The solar system case is validated against the JPL Horizons DE441 ephemeris. The galactic case is validated against the first-passage timing prediction of van der Marel et al. [11]. The Barnes–Hut force kernel is validated independently against the direct kernel on shared initial conditions.

Epoch-aligned comparison. For the solar system case, both the simulator and the JPL fetch use identical output intervals, sourced from the same scenario configuration. This avoids interpolation in the standard workflow, since both output streams sample at the same cadence from the same start epoch.

Relative position error. For each body at each epoch, the relative position error is defined as:

$$\epsilon_{\text{rel}}(t) = \frac{|\mathbf{r}_{\text{sim}}(t) - \mathbf{r}_{\text{JPL}}(t)|}{|\mathbf{r}_{\text{JPL}}(t)|} \quad (13)$$

where $|\cdot|$ denotes the Euclidean norm. For each body, we report the maximum of $\epsilon_{\text{rel}}(t)$ over all epochs

$t > 0$ (the *per-body maximum relative error*). The summary statistic reported in the results tables is the mean of these per-body maxima across the comparison set. This metric is informative for bodies whose barycentric distance is large relative to the position error, but can be misleading for bodies that remain close to the barycenter (notably the Sun); see Section 5.2.

Absolute position error and RMS metrics. To complement the maximum relative error, we also report the absolute position error and RMS statistics:

$$e_{\text{abs}}(t) = |\mathbf{r}_{\text{sim}}(t) - \mathbf{r}_{\text{JPL}}(t)| \quad (14)$$

$$\text{RMS}_{\text{rel}} = \sqrt{\frac{1}{N_t} \sum_t \epsilon_{\text{rel}}(t)^2} \quad (15)$$

$$\text{RMS}_{\text{abs}} = \sqrt{\frac{1}{N_t} \sum_t e_{\text{abs}}(t)^2} \quad (16)$$

These supplemental metrics are particularly important for bodies whose barycentric radius is small, for which the normalization in Eq. (13) can inflate the reported percentage. The RMS metrics characterize typical disagreement over the full integration, in contrast to the maximum which may be dominated by a single epoch.

Galactic first-passage timing. For the MW–M31 collision, the validation observable is the time of first close passage, defined as the first local minimum of the inter-galaxy separation $|\mathbf{R}_{\text{MW}}(t) - \mathbf{R}_{\text{M31}}(t)|$ after $t = 0$, where $\mathbf{R}$ denotes the COM of each half of the particle set. The reference value of 3.9 Gyr is taken from van der Marel et al. [11], which carries published uncertainty of order 10–15% driven principally by the M31 proper-motion measurement.

Energy and momentum conservation. Total mechanical energy, angular momentum, and linear momentum are sampled at regular intervals through-

out the integration:

$$T = \sum_{i=1}^N \frac{1}{2} m_i |\mathbf{v}_i|^2 \quad (17)$$

$$U = -G \sum_{i=1}^N \sum_{j=i+1}^N \frac{m_i m_j}{\sqrt{|\mathbf{r}_j - \mathbf{r}_i|^2 + \varepsilon^2}} \quad (18)$$

$$\mathbf{L} = \sum_{i=1}^N m_i \mathbf{r}_i \times \mathbf{v}_i, \quad \mathbf{P} = \sum_{i=1}^N m_i \mathbf{v}_i. \quad (19)$$

Drift is quantified by the relative peak-to-peak variation for energy,

$$\frac{\Delta E}{E} = \frac{|E_{\text{max}} - E_{\text{min}}|}{|E_0|}, \quad (20)$$

and by the maximum vector drift, normalized to the initial magnitude, for angular and linear momentum,

$$\frac{\Delta L}{L} = \frac{\max_t |\mathbf{L}(t) - \mathbf{L}_0|}{|\mathbf{L}_0|}, \quad (21)$$

$$\frac{\Delta P}{P} = \frac{\max_t |\mathbf{P}(t) - \mathbf{P}_0|}{|\mathbf{P}_0|}. \quad (22)$$

The vector-based definition for $\mathbf{L}$ and $\mathbf{P}$ captures both magnitude changes and directional rotation of the vector, unlike a scalar magnitude comparison which can miss rotational drift. Note that the peak-to-peak measure of Eq. (20) can differ by up to a factor of 2 from the more commonly reported $\max_t |E(t) - E_0|/|E_0|$ [2, 14]. The peak-to-peak form is a conservative upper bound. For the solar system case, the initial total momentum is generally nonzero because the selected 35-body subset is not the complete solar system; observed variation in $\mathbf{P}$ is therefore interpreted as numerical error rather than physical evolution. For the galactic case the COM of the full system is subtracted at $t = 0$ (Section 3.5.2), making $|\mathbf{P}_0|$ vanishing to machine precision, so the relative metric of Eq. (22) is uninformative there; absolute drift in $\mathbf{P}$ is monitored instead.

5. Solar System Results

5.1. 100-Year Simulation

Table 1 summarizes results for a 100-year integration (1950–2050) with $\Delta t = 360$ s, the Yoshida integrator, and the direct force kernel. The simulation completes in 56.5 s on a single core.

Table 1: 100-year simulation summary (35 bodies, $\Delta t = 360$ s, Yoshida 4th-order, direct kernel).

Metric	Value
Mean max. relative error [†]	0.053%
Worst-case body [†]	Ganymede (0.290%)
Planet error range	0.0003%–0.173%
Moon error range	0.009%–0.290%
Energy conservation $\Delta E/E$	7.4×10^{-13}
Angular momentum $\Delta L/L$	3.4×10^{-14}
Linear momentum $\Delta P/P$	1.0×10^{-8}
Wall-clock runtime	56.5 s

[†]Sun excluded; see Section 5.2.

Table 2 presents the per-body breakdown, grouped by dynamical class.

Table 2: Per-body maximum relative position error (%) for the 100-year simulation.

Body	Error (%)	Body	Error (%)
<i>Sun and Planets</i>			
Saturn	0.0003	Venus	0.022
Uranus	0.0003	Sun	0.071
Neptune	0.0004	Mercury	0.173
Jupiter	0.0009		
Mars	0.006		
Pluto	0.010		
Earth	0.013		
<i>Selected Moons</i>			
Iapetus	0.009	Io	0.114
Miranda	0.010	Titan	0.125
Phobos	0.014	Europa	0.175
Triton	0.016	Callisto	0.188
Moon	0.018	Ganymede	0.290

5.2. 249-Year Simulation

Extending the integration to 249 years (1950–2199) tests the long-term stability of the symplectic integrator. Table 3 compares the two simulation durations, and Table 4 presents the per-body breakdown.

Table 3: Comparison of 100-year and 249-year simulations ($\Delta t = 360$ s, direct kernel).

Metric	100 yr	249 yr
Mean max. error [†]	0.053%	0.079%
Worst body [†]	Ganymede (0.29%)	Callisto (0.41%)
Worst planet	Mercury (0.17%)	Mercury (0.42%)
$\Delta E/E$	7.4×10^{-13}	1.5×10^{-12}
$\Delta L/L$	3.4×10^{-14}	4.3×10^{-13}
$\Delta P/P$	1.0×10^{-8}	7.0×10^{-8}
Runtime	56.5 s	171.6 s

[†]Sun excluded; see Section 5.2.

Table 4: Per-body position error (%) for the 249-year simulation.

Body	Max Rel (%)	RMS Rel (%)
<i>Sun and Planets</i>		
Saturn	0.0007	0.0004
Uranus	0.0007	0.0004
Neptune	0.0009	0.0005
Jupiter	0.002	0.001
Mars	0.015	0.007
Pluto	0.034	0.016
Earth	0.031	0.018
Venus	0.055	0.031
Mercury	0.417	0.160
Sun	1.336	0.049
<i>Selected Moons</i>		
Miranda	0.010	0.006
Triton	0.016	0.011
Umbriel	0.019	0.013
Iapetus	0.022	0.012
Phobos	0.022	0.009
Moon	0.045	0.019
Amalthea	0.050	0.033
Io	0.116	0.077
Nereid	0.149	0.052
Europa	0.184	0.122
Titan	0.184	0.133
Hyperion	0.226	0.153
Ganymede	0.290	0.194
Callisto	0.411	0.246

Several observations follow from these results.

Energy and angular momentum conservation remain bounded. The energy drift increases only from 7.4×10^{-13} to 1.5×10^{-12} when the integration duration grows by a factor of 2.5. This sub-linear growth is characteristic of symplectic integrators, which approximately preserve a shadow

Hamiltonian $\tilde{H}$ and exhibit only oscillatory energy variations, never secular drift. Angular momentum conservation follows the same pattern, with $\Delta L/L$ remaining at $\sim 10^{-13}$ over 249 years.

Linear momentum conservation confirms integrator correctness. Because the simulated system is closed under its own pairwise Newtonian interactions, total linear momentum should be exactly conserved up to numerical error, even though $\mathbf{P}_0 \neq 0$. The observed drift of $\Delta P/P \sim 10^{-8}$ is consistent with accumulated floating-point roundoff over $\sim 2.2 \times 10^7$ timesteps, confirming that the integrator correctly preserves this conservation law. The growth from 1.0×10^{-8} to 7.0×10^{-8} over $2.5\times$ longer integration is consistent with sub-linear roundoff accumulation.

Positional errors grow sub-linearly. The mean error (excluding the Sun) grows from 0.053% to 0.079% over $2.5\times$ the integration time. This sub-linear growth is consistent with slow phase drift accumulation, not exponential divergence. The outer planets (Saturn, Uranus, Neptune) remain below $10^{-3}\%$ even at 249 years, reflecting the dominance of solar gravitation in their dynamics and the absence of significant missing physics at their orbital distances.

The Sun’s large relative error is driven by coordinate normalization. Although the Sun attains the largest maximum relative error (1.336%) at 249 years, the comparison is performed in solar-system barycentric coordinates and normalized by the instantaneous JPL barycentric radius. The Sun’s barycentric displacement ($\sim 10^6$ km) is three orders of magnitude smaller than that of the outer planets ($\sim 10^9$ km), so an absolute position error of order 10^2 km inflates to a percentage that is not comparable across bodies. The effect is further amplified by reporting the maximum over all epochs, so a single small-denominator epoch can dominate. Absolute-error metrics resolve the ambiguity: the Sun’s maximum absolute position error is 238 km with RMS absolute error 159 km, both smaller than those of

any other body in the full 35-body comparison. Its RMS relative error is 0.049%, which is far more representative of its overall agreement with JPL. The Sun is therefore excluded from the summary mean reported in Tables 1 and 3.

Callisto overtakes Ganymede at 249 years. Callisto’s error grows from 0.188% to 0.411%, surpassing Ganymede (0.290% at both durations). This is consistent with the longer orbital period of Callisto (16.7 d vs. 7.15 d), which couples to slightly different phase drift accumulation rates over multi-century timescales. Hyperion (0.226%) also emerges as a notable outlier among Saturnian moons, likely reflecting its sensitivity to perturbations from Titan and Saturn’s other satellites, combined with its eccentric and inclined orbit.

Absolute position error. Table 5 supplements the relative metric with absolute position error in km for selected bodies in the 249-year simulation. This provides a normalization-free assessment of positional disagreement that is not distorted by the small barycentric radius of the Sun.

Table 5: Absolute position error (km) for the 249-year simulation, selected bodies.

Body	Max (km)	RMS (km)
<i>Sun and Planets</i>		
Sun	238	159
Mercury	194,898	86,793
Venus	59,238	33,903
Earth	46,554	26,303
Mars	29,946	15,532
Jupiter	16,180	9,026
Saturn	9,627	5,340
<i>Selected Moons</i>		
Callisto	3,111,029	1,912,082
Ganymede	2,153,442	1,504,302
Titan	2,494,262	1,903,692
Io	858,963	597,530
Moon	65,996	28,728

The Sun’s maximum absolute error of 238 km over 249 years is the smallest of any body in the full 35-body comparison (Table 5 shows a representative subset), confirming that the 1.34% relative figure is a

consequence of normalization rather than dynamical disagreement. Among planets, the error hierarchy tracks the expected importance of missing physics: Mercury (relativistic precession), then Venus and Earth (asteroid perturbations, J_2), then outer giants (minimal missing effects). The large absolute errors for Galilean moons and Titan reflect accumulated phase drift over tens of thousands of fast orbits, not growing instability.

5.3. Sources of Residual Error

The residual errors are dominated by physics model differences between the simulator (Newtonian gravity, 35 bodies) and JPL DE441. The principal missing contributions are described below.

Post-Newtonian relativity. The Einstein–Infeld–Hoffmann (EIH) equations contribute ~ 43 arcsec century $^{-1}$ of precession to Mercury’s orbit [12]. This is the dominant error source for Mercury (0.17% at 100 yr, 0.42% at 249 yr), as the Newtonian model cannot capture this effect. Implementation of 1PN corrections was attempted; however, the velocity-dependent post-Newtonian potential breaks the separability assumption ($H = T(\mathbf{p}) + V(\mathbf{q})$) required by the Yoshida integrator. This degraded energy conservation by ~ 4 orders of magnitude (from 10^{-13} to 10^{-9}), producing worse overall accuracy than the purely Newtonian model. Implicit symplectic splitting schemes [13] or mixed-variable symplectic methods [12] offer potential paths to incorporating these corrections without sacrificing symplecticity.

Solar oblateness (J_2). The Sun’s quadrupole moment produces a small additional precession, particularly affecting Mercury’s perihelion advance.

Asteroid perturbations. JPL DE441 includes gravitational contributions from over 300 asteroids (principally Ceres, Pallas, and Vesta). These bodies are absent from the 35-body model.

Moon phase drift. Satellites with short orbital periods accumulate phase errors over many orbits. Io orbits Jupiter every 1.77 d and completes $\sim 20,600$

orbits in 100 years. At $\Delta t = 360$ s, each orbit is resolved by ~ 425 timesteps. This is adequate for 4th-order per-orbit accuracy, but small per-orbit phase errors compound over thousands of orbits. Reducing Δt or implementing adaptive timestepping would improve satellite accuracy at the cost of runtime.

Tidal and non-gravitational effects. Tidal dissipation, solar radiation pressure, and planetary ring interactions are included in DE441 but absent from the simulator.

5.4. Timestep Convergence Study

To distinguish integration error from physics model error, a systematic convergence study was performed over a range of timesteps spanning two orders of magnitude. Table 6 presents the peak-to-peak energy drift $\Delta E/E$ for the 249-year simulation at each Δt .

Table 6: Energy conservation vs. timestep (249-year simulation, 35 bodies, Yoshida 4th-order, direct kernel).

Δt (s)	$\Delta E/E$
60	3.74×10^{-12}
180	3.20×10^{-12}
360	1.46×10^{-12}
600	1.22×10^{-12}
900	2.08×10^{-12}
1800	2.40×10^{-11}
3600	4.79×10^{-10}
7200	4.34×10^{-8}
9600	7.55×10^{-9}

The data reveal three distinct regimes of integrator behavior.

Regime A: Precision floor ($\Delta t \leq 900$ s). For Δt from 60 s to 900 s, the energy drift remains at $\sim 10^{-12}$ with no clear 4th-order scaling. The minimum occurs near $\Delta t = 600$ s (1.22×10^{-12}); at smaller timesteps, the drift increases mildly (to 3.74×10^{-12} at $\Delta t = 60$ s), consistent with accumulated floating-point roundoff over a larger number of arithmetic operations. This floor arises from finite-precision arithmetic in the force summation and long-time energy evaluation, not from the inte-

grator itself. The production timestep $\Delta t = 360$ s is well within this converged regime.

Regime B: 4th-order convergence ($900 \leq \Delta t \leq 3600$ s). In this range, the energy error scales approximately as $\mathcal{O}(\Delta t^4)$, consistent with the expected behavior of the Yoshida integrator. From $\Delta t = 900$ s to 1800 s (a factor of 2), the energy drift increases by a factor of ~ 11.5 (expected: $2^4 = 16$). From 1800 s to 3600 s, the drift increases by a factor of ~ 20 (expected: 16). Both ratios are in reasonable agreement with 4th-order scaling; the deviation from the theoretical factor of 16 is expected when measuring peak-to-peak variation over a long integration rather than single-step local truncation error.

Regime C: Dynamical under-resolution ($\Delta t \geq 7200$ s). At $\Delta t = 7200$ s (~ 2 hr), the energy drift exceeds the 4th-order prediction by a large factor: the ratio from 3600 s to 7200 s is ~ 90 , far exceeding the expected 16. At this timestep, the fastest-orbiting body (Io, period 1.77 d) is resolved by only ~ 21 timesteps per orbit. The symplectic splitting approximation degrades when the timestep becomes comparable to the dynamical timescale of the system. The non-monotonic behavior at $\Delta t = 9600$ s ($\Delta E/E$ decreasing relative to 7200 s) is consistent with the oscillatory energy error characteristic of symplectic integrators: the peak-to-peak variation depends on the resonance between the timestep and the orbital frequencies, and can fluctuate non-monotonically at large Δt [2].

Implications. The convergence study strongly indicates that residual positional errors are dominated by model discrepancy rather than integrator truncation error. In Regime A, the integration error has converged to the floating-point floor, yet positional errors against JPL remain unchanged. Furthermore, the error hierarchy matches the expected importance of missing physics. Any timestep within Regime A ($\Delta t \lesssim 900$ s) produces indistinguishable energy conservation; the production timestep $\Delta t = 360$ s is adopted for a modest additional margin.

6. Galactic Collision Results

A production run of the MW–M31 collision is performed at $N = 50,000$ macro-particles ($N/2$ per galaxy) with the Yoshida 4th-order integrator, the Barnes–Hut force kernel at opening parameter $\theta = 0.5$, timestep $\Delta t = 1$ Myr, total duration 6 Gyr, and softening $\varepsilon = 100$ pc. Initial conditions follow the van der Marel et al. [11] parameters of Section 3.5.2. The simulation completes in 52.7 minutes on a 12-thread Zen 4 host. Table 7 summarizes the results.

Table 7: MW–M31 collision summary ($N = 50,000$, $\Delta t = 1$ Myr, 6 Gyr, Yoshida, BH $\theta = 0.5$, $\varepsilon = 100$ pc).

Quantity	Value
Galaxy mass (each)	$1.5 \times 10^{12} M_{\odot}$
Plummer scale radius a	30 kpc
Initial separation (recovered)	769.8 kpc
Initial radial velocity	109 km s^{-1} (closing)
Initial tangential velocity	17 km s^{-1}
First close passage (this work)	3.72 Gyr
First close passage (vdM 2012)	3.9 Gyr
Deviation	4.6%
Energy drift $\Delta E/E$	1.44%
Angular momentum drift $\Delta L/L$	0.19%
Wall-clock runtime	52.7 min

6.1. First-passage timing

The simulated first close passage occurs at 3.72 Gyr, compared with the van der Marel et al. [11] predicted value of 3.9 Gyr. This is a 4.6% deviation, comfortably within the 10–15% uncertainty band that the literature ascribes to the M31 tangential velocity measurement, which is the principal driver of the predicted passage time. The recovered initial separation of 769.8 kpc agrees with the configured value of 770 kpc to within the precision of the Plummer-sphere sampling; this confirms that the COM subtraction of Section 3.5.2 positions the two galaxies as intended. The inter-galaxy separation as a function of time is reported in the supplementary plot `tests/galaxy_separation.png` generated by `tools/galaxy/compare.py`.

6.2. Conservation diagnostics

The energy drift of 1.44% over 6 Gyr is many orders of magnitude larger than the 10^{-12} achieved in the solar system integration, and warrants discussion. Three effects are at play. First, the timestep is much larger relative to the dynamical timescales of individual particles: at the Plummer core, the orbital period is of order 10–100 Myr, so $\Delta t = 1$ Myr resolves the fastest orbits with only $\mathcal{O}(10\text{--}100)$ steps per orbit, compared to ~ 425 for Io in the solar case. Second, the Barnes–Hut multipole approximation introduces a force error of order θ^2 per cell, which manifests as a small non-conservative perturbation that the symplectic integrator cannot absorb into its shadow Hamiltonian. Third, particles undergoing close passes within the softening length experience a smoothed potential rather than the bare $1/r$, so the energy book-keeping over close encounters is approximate. The 1.44% drift is consistent with standard practice in collisionless galactic N-body work at this resolution and is well within the regime where bulk dynamics (orbits, passage times, tidal structure) are not compromised.

The angular momentum drift of 0.19% is more than two orders of magnitude smaller than the energy drift, indicating that the integrator preserves the rotational symmetry of the problem well even when the force evaluation is approximate. The relative linear momentum drift is uninformative here, since $|\mathbf{P}_0|$ vanishes by construction; the absolute drift in $|\mathbf{P}|$ remains at the floating-point roundoff floor.

6.3. Limitations

The galactic configuration is a deliberate simplification of the real system, intended as a Barnes–Hut validation target rather than a cosmological prediction. The principal limitations are:

Single-component Plummer profile. Each galaxy is modeled as a single isotropic Plummer sphere, not as a separated disc, bulge, and dark matter halo. The Plummer density falls as r^{-5} at large radius, whereas the Hernquist profile [7], which is

closer to observed elliptical-galaxy and halo profiles, falls as r^{-4} . The Plummer choice is standard for analytic tractability of the self-consistent velocity sampling (Eq. 12) but is known to under-predict mass at large radii.

No live cosmological context. The simulation has no Hubble flow, no dark matter halo coupling to the cosmic web, and no gas or stellar physics. It evolves two isolated self-gravitating spheres in free space.

Coarse macro-particle sampling. $N = 50,000$ per pair is modest by modern standards. Two-body relaxation per crossing time scales as $N/(8 \ln N)$, so artificial scattering is non-negligible. The 100 pc softening suppresses but does not eliminate this effect.

These limitations affect quantitative observables (the precise merger timeline beyond first passage, the morphology of tidal debris) but do not invalidate the first-passage timing comparison, which is governed primarily by the initial bulk phase-space configuration of the two galaxies and is robust to the internal density profile.

7. Barnes–Hut Validation

The Barnes–Hut kernel is verified through a dedicated unit-test suite (8 tests under `tests/unit_tests/barnes_hut.cpp`, part of a 23-test overall suite). Tests are registered across multiple translation units via an inline-global Meyers-singleton pattern.

7.1. Correctness tests

The following invariants are exercised on each build:

- **Self-interaction.** A single particle in isolation has zero gravitational acceleration. This tests that the leaf code correctly skips the self-pair.
- **Two-body match against direct.** For $N = 2$, the BH kernel must reproduce the direct kernel exactly (up to floating-point roundoff); there is no multipole approximation possible at $N = 2$.
- **Small cluster at low θ .** For $\theta \rightarrow 0$, the BH tree never accepts an internal node and effectively

degenerates to direct summation. Forces must match the direct kernel to within roundoff.

- **Kepler closure.** A two-body Kepler orbit integrated over one period must return to its initial state, confirming that no spurious dissipation is introduced by the tree-based force.
- **Energy drift bounded.** On a uniform-cube distribution, energy drift over a fixed integration interval is below a configured tolerance.
- **Plummer accuracy.** On a Plummer-sphere initial condition, the mean relative force error from BH against direct is below a tolerance set by the test θ .
- **Uniform cube accuracy.** As above, on a uniform random distribution in a cube.
- **Monotonic θ improvement.** For a sweep over decreasing θ on a fixed distribution, the mean relative force error must decrease monotonically.

7.2. Theta-error scaling

The monotonic-improvement test deserves a remark on what it does *not* assert. Per cell, the leading correction to the multipole expansion about the COM is quadrupole, giving relative force error $\mathcal{O}((s/d)^2) = \mathcal{O}(\theta^2)$. The aggregate mean relative force error across all particles, however, is observed empirically to scale somewhere between θ^3 and θ^4 depending on the distribution and the range of θ examined. This is consistent with prior observations [6]: averaging over many cells with locally symmetric mass distributions produces partial cancellations of the quadrupole error, yielding aggregate scaling steeper than the per-cell exponent. The unit test asserts only monotonic improvement, which is the property mathematically guaranteed by the algorithm; the precise scaling exponent is reported separately in benchmarks but not enforced as a correctness criterion.

8. Performance

8.1. Direct kernel

The direct force computation scales as $\mathcal{O}(N^2)$ per timestep due to pairwise summation. The Yoshida integrator requires 3 force evaluations per step, yielding a total cost of $\mathcal{O}(3N^2 \cdot T/\Delta t)$ for a simulation of duration T . For the 35-body solar system with $\Delta t = 360$ s over 249 years (Julian year of 365.25 d), this is $3 \times 35^2 \times 21,827,340 \approx 8.0 \times 10^{10}$ pairwise interaction evaluations, completed in ~ 172 s on a single core.

The outer loop is parallelized with OpenMP for $N \geq 350$; below that threshold the thread-management overhead exceeds the parallel benefit and OpenMP is bypassed. SIMD vectorization of the inner loop is active at all N .

8.2. Barnes–Hut versus direct

A scaling benchmark sweeps N from 512 to 131,072 on a 12-thread Zen 4 host (AMD Ryzen 5 8500G), with $\Delta t = 900$ s, the Yoshida integrator, and BH opening parameter $\theta = 0.5$. Both kernels are run with SIMD and OpenMP parallelism enabled. The benchmark auto-calibrates the step count per configuration so that each trial of the slower kernel takes approximately 2 s, and reports the median wall time over three trials. Serial-direct timings are skipped at this N range as the parallel direct kernel is the meaningful baseline. Table 8 reports the resulting per-step times.

Table 8: Per-step wall time (ms) for direct vs. Barnes–Hut on Ryzen 5 8500G with 12 OpenMP threads, $\Delta t = 900$ s, $\theta = 0.5$. Median of 3 trials.

N	Direct (ms)	BH (ms)	Speedup
512	0.717	0.652	1.10×
1,024	2.450	1.369	1.79×
2,048	9.421	3.774	2.50×
4,096	35.978	9.067	3.97×
8,192	142.893	15.121	9.45×
16,384	557.000	42.767	13.02×
32,768	2,217.2	124.533	17.80×
65,536	9,219.6	210.167	43.87×
131,072	35,597.5	676.833	52.59×

Two observations are noteworthy.

Crossover at or below $N = 512$. On this host the Barnes–Hut kernel is faster than the parallel direct kernel at every measured $N \geq 512$, with a minimum speedup of $1.10\times$ at the smallest measured size. The true crossover lies at or below 512, which is earlier than the $N \in [10^3, 10^4]$ band often cited in textbook treatments of Barnes–Hut. Two factors contribute: the direct kernel has been aggressively optimized (SIMD, OpenMP, SoA, branchless masking, restrict-qualified pointers), so it represents a strong baseline; and the BH kernel benefits from cache-friendly access through the leaf index array, keeping per-cell evaluation cheap even at modest N .

Speedup grows with N as expected for $\mathcal{O}(N \log N)$ vs. $\mathcal{O}(N^2)$. The speedup increases from $1.10\times$ at $N = 512$ to $52.59\times$ at $N = 131,072$. The ratio of asymptotic costs scales as $N/\log_2 N$. From $N = 512$ to $N = 131,072$ this predicts a speedup ratio of $(131072/17)/(512/9) = 7710/56.9 \approx 135$, in reasonable order-of-magnitude agreement with the observed factor of $52.59/1.10 = 47.8$. The shortfall against the asymptotic prediction reflects two effects: fixed per-step overheads of the BH tree (allocation, COM aggregation, traversal stack setup) that do not scale with $N \log N$, and constant-factor inefficiencies in tree traversal (branch mispredictions, irregular memory access) that the direct kernel’s regular SIMD inner loop avoids. The observed direct timings track N^2 closely: from $N = 16,384$ to $N = 131,072$ (an $8\times$ increase in N) direct time grows by $35597/557 \approx 64\times$, matching $8^2 = 64$ within rounding.

8.3. Memory layout

The SoA layout described in Section 3.2 stores 13 scalar fields per particle. All sub-arrays occupy a single contiguous allocation with each sub-array start aligned to the SIMD register width and padded to a SIMD-element boundary, so the total allocation is $13 \times \lceil N/w \rceil \times w$ doubles, where w is the SIMD width in elements. For the 35-body solar configuration

the entire particle dataset fits in L1 cache. For the $N = 50,000$ galactic configuration the dataset is ~ 5 MB and fits in L2; at $N = 131,072$ in the scaling benchmark it occupies ~ 13 MB and spills to L3. The `_restrict_` qualifiers permit the compiler to assume no pointer aliasing, enabling store-to-load forwarding optimizations.

9. Conclusion

We have presented a symplectic N-body simulator that supports two complementary force kernels under a single integration framework. The 4th-order Yoshida integrator achieves energy conservation approaching the double-precision roundoff floor ($\Delta E/E \leq 10^{-12}$) and angular momentum conservation at $\Delta L/L \leq 10^{-13}$ over multi-century integration of the solar system, with positional accuracy of 0.079% mean per-body maximum relative error against JPL Horizons DE441 over 249 years across 34 non-Sun bodies. The dominant residual errors are consistent with missing physics in the Newtonian 35-body model, particularly relativistic precession for Mercury and long-horizon phase drift for fast-orbiting satellites. A systematic timestep convergence study identifies $\Delta t \lesssim 900$ s as the converged regime where integration error has reached the floating-point precision floor.

The Barnes–Hut octree kernel uses an explicit leaf discriminant in place of the brittle child-pointer sentinel convention, and applies the multipole acceptance criterion using geometric box center for separation distance and center of mass for force accumulation. This yields quadrupole-leading per-cell error and aggregate empirical error scaling between θ^3 and θ^4 . On a 12-thread Zen 4 host the tree kernel outperforms the SIMD- and OpenMP-vectorized direct kernel at every measured $N \geq 512$, with a $52.6\times$ speedup at $N = 131,072$.

The galactic capability is demonstrated on a Milky Way–M31 close-encounter scenario configured with the parameters of van der Marel et al. (2012). At $N = 50,000$ over 6 Gyr the simulation reproduces the

predicted first close passage at 3.72 Gyr against the literature value of 3.9 Gyr, a 4.6% deviation within the published uncertainty band.

Future work includes implementing post-Newtonian corrections via an implicit or mixed-variable symplectic scheme, adding solar oblateness perturbations, and porting both force kernels to CUDA with shared-memory tiling for the direct kernel and a Karras LBVH construction for the tree.

Code Availability

The source code for the simulator and the validation pipeline are open source and available at <https://github.com/karl-kes/N-Body-Physics-Engine>.

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